

Literature Status: Hardy-Space Quasi-Greedy Democracy

Automatic Math Discovery Run `fa.banach.001`

June 14, 2026

Original Problem

The source paper is Fernando Albiac, Jose L. Ansorena, and Przemyslaw Wojtaszczyk, *Quasi-greedy bases in the spaces ℓ_p ($0 < p < 1$) are democratic*, arXiv:2004.05206; later published in *J. Funct. Anal.* 280 (2021), no. 7, 108871.

Question 3.8 appears on PDF page 16 of the source paper, after the proof that quasi-greedy bases in ℓ_p , $0 < p < 1$, are democratic. It asks:

Let $0 < p < 1$ and $d \in \mathbb{N}$. Are all quasi-greedy bases in $H_p(\mathbb{D}^d)$ democratic?

The same passage is in `data/parsed/arxiv_sources/2004.05206/source.tex`, lines 555–557.

Supporting Answer Source

The supporting paper is Fernando Albiac, Jose L. Ansorena, and Glenier Bello, *Democracy of quasi-greedy bases in p -Banach spaces with applications to the efficiency of the TGA in the Hardy spaces $H_p(\mathbb{D}^d)$* , arXiv:2208.09342.

Its abstract says that the authors prove all quasi-greedy bases of $H_p(\mathbb{D})$, $0 < p < 1$, are democratic, while no quasi-greedy basis of $H_p(\mathbb{D}^d)$ for $d \geq 2$ is democratic, and that this solves a problem raised in [1]. In the introduction, PDF page 4, the paper states explicitly that it solves Question 3.8 from [1] positively when $d = 1$, and proves that no quasi-greedy basis of the multivariate Hardy spaces is democratic when $d \geq 2$. Its bibliography identifies [1] as the JFA version of arXiv:2004.05206.

Answer

Question 3.8 from arXiv:2004.05206 is already answered in the later literature. The answer depends on the dimension:

$d = 1$: yes, all quasi-greedy bases of $H_p(\mathbb{D})$ are democratic;

$d \geq 2$: no, no quasi-greedy basis of $H_p(\mathbb{D}^d)$ is democratic.

For $d = 1$, Corollary 4.3 of arXiv:2208.09342, PDF page 21, proves more generally that if $\mathcal{Y}$ is a truncation quasi-greedy basis of a complemented subspace of $H_p(\mathbb{D})$, then $\mathcal{Y}$ is democratic with upper democracy function equivalent to $m^{1/p}$. In particular, all quasi-greedy bases of $H_p(\mathbb{D})$ are almost greedy, hence democratic.

For $d \geq 2$, Corollary 4.6 of arXiv:2208.09342, PDF page 22, proves that $H_p(\mathbb{D}^d)$ has no squeeze-symmetric bases and in particular no almost greedy bases. Together with the quasi-Banach greedy-basis characterization used by the authors, this gives the negative answer stated explicitly in their abstract and introduction: no quasi-greedy basis of $H_p(\mathbb{D}^d)$, $d \geq 2$, is democratic.

Verification Notes

Local source text checks:

- `data/parsed/arxiv_sources/2004.05206/source.tex`, lines 555–557, contains the source Question 3.8.
- `data/parsed/arxiv_sources/2208.09342/source.tex`, line 114, states in the abstract that the paper solves the problem raised in [1].
- `data/parsed/arxiv_sources/2208.09342/source.tex`, line 142, identifies the source question as Question 3.8 and states the $d = 1$ and $d \geq 2$ outcomes.
- `data/parsed/arxiv_sources/2208.09342/source.tex`, lines 726–733, contains Corollary 4.3 for $H_p(\mathbb{D})$.
- `data/parsed/arxiv_sources/2208.09342/source.tex`, lines 779–784, contains Corollary 4.6 for $H_p(\mathbb{D}^d)$, $d \geq 2$.
- `data/parsed/arxiv_sources/2208.09342/source.tex`, lines 944–956, identifies [1] as the JFA paper corresponding to arXiv:2004.05206.

Search Bounds

Before making this packet, the run registry, solution index, attempt index, and proof-gap index were searched for arXiv:2004.05206, arXiv:2208.09342, “Hardy”, H_p , $H_p(\mathbb{D}^d)$, and quasi-greedy democracy terms. No existing packet or ledger row for this exact literature answer was found. The current target pool already contained arXiv:2208.09342 as a related later paper. A web search on June 14, 2026 found the arXiv record for arXiv:2208.09342, whose abstract gives the same explicit resolution.

Limitations

This packet records an already-known literature answer, not an original proof or counterexample produced by the run. It does not claim any resolution of new questions raised in arXiv:2208.09342.

Human Review Recommendation

Check the bibliographic identification of [1] with arXiv:2004.05206 and confirm that the source Question 3.8 is the same Hardy-space democracy question answered in arXiv:2208.09342. The supporting paper explicitly identifies the problem, so review should be straightforward.

References

- [1] F. Albiac, J. L. Ansorena, and P. Wojtaszczyk, *Quasi-greedy bases in the spaces ℓ_p ($0 < p < 1$) are democratic*, J. Funct. Anal. 280 (2021), no. 7, 108871; arXiv:2004.05206.
- [2] F. Albiac, J. L. Ansorena, and G. Bello, *Democracy of quasi-greedy bases in p -Banach spaces with applications to the efficiency of the TGA in the Hardy spaces $H_p(\mathbb{D}^d)$* , arXiv:2208.09342.